Lab 7: Fuzz Testing with Hypothesis

SRN: PES1UG24AM385
Project: L6_L7_40 Ride Share Fare

Step 1: Install dependencies

Hypothesis and pytest were installed with pip. Hypothesis is pinned to 6.112.0 because 6.168.5 crashed with `AttributeError: 'TreeNode' object has no attribute 'is_exhausted'`.

```
python -m pip install --force-reinstall --no-cache-dir "hypothesis==6.112.0"
pytest
```

Versions used for all runs in this report: Python 3.10.20, pytest 9.2.0, hypothesis 6.112.0.

Step 2: Run the buggy application

With well-formed input the original program works and prints the expected results.

```
PS C:\Users\deepg\Downloads\SELAB6> python ride_share_fare.py
==== Running parse_trip ====
Enter cab:distance in km (e.g. sedan:12): sedan:12
Cab and Distance: ('Sedan', 12.0)

==== Running fare_average ====
Enter fares as CSV (e.g. 120,95,210): 120,95,210
Average Fare: 141

==== Running payable_fare ====
Enter fare|promo (e.g. 250|30): 250|30
Payable Fare: 265.0
```

Buggy version, valid input (sedan:12 / 120,95,210 / 250|30)

With a malformed input (`sedan`, no `:` separator) the program crashes with an unhandled `ValueError`.

```
PS C:\Users\deepg\Downloads\SELAB6> python ride_share_fare.py
==== Running parse_trip ====
Enter cab:distance in km (e.g. sedan:12): sedan
Traceback (most recent call last):
  File "/home/user/SELAB6/ride_share_fare.py", line 31, in <module>
    main()
  File "/home/user/SELAB6/ride_share_fare.py", line 17, in main
    result = parse_trip(user_input)
  File "/home/user/SELAB6/ride_share_fare.py", line 2, in parse_trip
    cab, km = text.split(':')
ValueError: not enough values to unpack (expected 2, got 1)
```

Buggy version, invalid input 'sedan' - crash with traceback

Steps 3-4: Write and run the Hypothesis fuzz tests (failing)

`test_ride_share_fare.py` has one property-based test per function (500 examples each). The strategies mix random text, `None`, integers and almost-valid strings (numbers, nan, inf, empty parts, extra separators). Each test asserts that the function never raises and that any returned value is sensible (non-empty cab name, finite non-negative numbers). Against the buggy code all three tests fail.

```
=========================== short test summary info ============================
FAILED test_ride_share_fare.py::test_parse_trip_no_crash - exceptiongroup.Exc...
FAILED test_ride_share_fare.py::test_fare_average_no_crash - exceptiongroup.E...
FAILED test_ride_share_fare.py::test_payable_fare_no_crash - exceptiongroup.E...
============================== 3 failed in 5.42s ===============================
```

pytest summary on the buggy version: 3 failed

```
PS C:\Users\deepg\Downloads\SELAB6> python -m pytest test_ride_share_fare.py
============================= test session starts ==============================
platform linux -- Python 3.10.20, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/user/SELAB6
plugins: hypothesis-6.112.0
collected 3 items

test_ride_share_fare.py FFF                                              [100%]

================================== FAILURES ===================================
_____________________________ test_parse_trip_no_crash ______________________________
  + Exception Group Traceback (most recent call last):
  |   File "/home/user/SELAB6/test_ride_share_fare.py", line 37, in test_parse_trip_no_crash
  |     @given(trip_inputs)
  | exceptiongroup.ExceptionGroup: Hypothesis found 5 distinct failures. (5 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/user/SELAB6/test_ride_share_fare.py", line 47, in test_parse_trip_no_crash
    |     assert km >= 0 and km != float("inf") and km == km
    | AssertionError: assert (nan >= 0)
    | Falsifying example: test_parse_trip_no_crash(
    |     s='0:nan',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/home/user/SELAB6/test_ride_share_fare.py", line 46, in test_parse_trip_no_crash
    |     assert isinstance(cab, str) and cab != ""
    | AssertionError: assert (True and '' != '')
    |  +  where True = isinstance('', str)
    | Falsifying example: test_parse_trip_no_crash(
    |     s=':nan',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/home/user/SELAB6/test_ride_share_fare.py", line 40, in test_parse_trip_no_crash
    |     result = parse_trip(s)
    |   File "/home/user/SELAB6/ride_share_fare.py", line 3, in parse_trip
    |     return cab.strip().title(), float(km)
    | ValueError: could not convert string to float: ''
    |
```

pytest output on the buggy version: start of run and the 5 distinct failures found in parse_trip (part 1 of 2)

```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/user/SELAB6/test_ride_share_fare.py", line 42, in test_parse_trip_no_crash
    |     assert False, f"parse_trip crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_trip crashed on ':' with: ValueError: could not convert string to float: ''
    | assert False
    | Falsifying example: test_parse_trip_no_crash(
    |     s=':',
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         /home/user/SELAB6/test_ride_share_fare.py:41
    +---------------- 4 ----------------
    | Traceback (most recent call last):
    |   File "/home/user/SELAB6/test_ride_share_fare.py", line 40, in test_parse_trip_no_crash
    |     result = parse_trip(s)
    |   File "/home/user/SELAB6/ride_share_fare.py", line 2, in parse_trip
    |     cab, km = text.split(':')
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/user/SELAB6/test_ride_share_fare.py", line 42, in test_parse_trip_no_crash
    |     assert False, f"parse_trip crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_trip crashed on '' with: ValueError: not enough values to unpack (expected 2, got 1)
    | assert False
    | Falsifying example: test_parse_trip_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         /home/user/SELAB6/test_ride_share_fare.py:41
    +---------------- 5 ----------------
    | Traceback (most recent call last):
    |   File "/home/user/SELAB6/test_ride_share_fare.py", line 40, in test_parse_trip_no_crash
    |     result = parse_trip(s)
    |   File "/home/user/SELAB6/ride_share_fare.py", line 2, in parse_trip
    |     cab, km = text.split(':')
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/user/SELAB6/test_ride_share_fare.py", line 42, in test_parse_trip_no_crash
    |     assert False, f"parse_trip crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_trip crashed on None with: AttributeError: 'NoneType' object has no attribute 'split'
    | assert False
    | Falsifying example: test_parse_trip_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         /home/user/SELAB6/test_ride_share_fare.py:41
    +------------------------------------
```

pytest output on the buggy version: start of run and the 5 distinct failures found in parse_trip (part 2 of 2)

Bugs found

Minimal failing inputs reported by Hypothesis (plus the manual crash from Step 2):

Function	Failing input	Error
`parse_trip`	`None`	AttributeError: 'NoneType' object has no attribute 'split'
`parse_trip`	`'' (empty)`	ValueError: not enough values to unpack (expected 2, got 1)
`parse_trip`	`'sedan'`	ValueError: not enough values to unpack (expected 2, got 1)
`parse_trip`	`':'`	ValueError: could not convert string to float: ''
`parse_trip`	`'0:nan'`	No crash, but nan accepted as distance (assert nan >= 0 fails)
`parse_trip`	`':nan'`	No crash, but empty cab name accepted (assert cab != '' fails)
`fare_average`	`None`	AttributeError: 'NoneType' object has no attribute 'split'
`fare_average`	`'' (empty)`	ValueError: invalid literal for int() with base 10: ''
`payable_fare`	`None`	AttributeError: 'NoneType' object has no attribute 'split'
`payable_fare`	`'' (empty)`	ValueError: not enough values to unpack (expected 2, got 1)

Step 5: Fix the application

The fix. Each function now checks that its argument is a string and that splitting gives exactly the expected number of parts. Numbers go through one helper, `_to_number()`, which rejects empty, non-numeric, non-finite (nan, inf, 1e400) and negative values. `parse_trip` also rejects an empty cab name, `fare_average` guards against an empty list, and `payable_fare` rejects an overflowing total. Invalid input returns `None` instead of raising, and `main()` prints an "Invalid input" message instead of a traceback.

Valid input still gives exactly the same results as before:

```
PS C:\Users\deepg\Downloads\SELAB6> python ride_share_fare.py
==== Running parse_trip ====
Enter cab:distance in km (e.g. sedan:12): sedan:12
Cab and Distance: ('Sedan', 12.0)

==== Running fare_average ====
Enter fares as CSV (e.g. 120,95,210): 120,95,210
Average Fare: 141

==== Running payable_fare ====
Enter fare|promo (e.g. 250|30): 250|30
Payable Fare: 265.0
```

Fixed version, valid input - same results as the original

Invalid input is now reported cleanly and the program keeps running:

```
PS C:\Users\deepg\Downloads\SELAB6> python ride_share_fare.py
==== Running parse_trip ====
Enter cab:distance in km (e.g. sedan:12): sedan
Invalid input: use the format cab:distance with a non-negative number

==== Running fare_average ====
Enter fares as CSV (e.g. 120,95,210): 12,,abc
Invalid input: enter non-negative whole numbers separated by commas

==== Running payable_fare ====
Enter fare|promo (e.g. 250|30): nan|-5
Invalid input: use the format fare|promo with non-negative numbers
```

Fixed version, invalid input (sedan / 12,,abc / nan|-5) - no crash

Step 6: Re-run the fuzz tests (passing)

The same test file, unchanged, run against the fixed code: all three property tests pass.

```
PS C:\Users\deepg\Downloads\SELAB6> python -m pytest test_ride_share_fare.py -v
============================= test session starts ==============================
platform linux -- Python 3.10.20, pytest-9.2.0, pluggy-1.7.0 -- /tmp/claude-0/-home-user/41c22517-6de8-5751-8782-47430
a6be59e/scratchpad/venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default' -> database=DirectoryBasedExampleDatabase(PosixPath('/home/user/SELAB6/.hypothesis/exampl
es'))
rootdir: /home/user/SELAB6
plugins: hypothesis-6.112.0
collecting ... collected 3 items

test_ride_share_fare.py::test_parse_trip_no_crash PASSED                 [ 33%]
test_ride_share_fare.py::test_fare_average_no_crash PASSED               [ 66%]
test_ride_share_fare.py::test_payable_fare_no_crash PASSED               [100%]

============================== 3 passed in 4.90s ===============================
```

pytest -v on the fixed version: 3 passed